Negative Self-Determination as a Source of Unhappiness: A Formal Theory Integrating Cognitive Appraisal, Predictive Processing, and Dynamical Systems

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Abstract

This paper develops a formal theory according to which unhappiness is not a primary affective datum but a derived cognitive state produced principally by a specific meta-cognitive operation termed *negative self-determination*. We distinguish rigorously between *suffering*—the first-order sensory, physiological, or situational distress that accompanies adverse events—and *unhappiness*—the second-order, reflexive judgment in which the subject identifies itself as "unhappy" or "defeated." Drawing on cognitive appraisal theory, Stoic philosophy, predictive processing frameworks, acceptance and commitment therapy, and dynamical-systems modelling, we formalise the claim that the cognitively mediated component of unhappiness arises through such a self-referential assertion. We introduce a minimal operator-theoretic notation, develop a refined three-component model decomposing the process into primary affective reaction, cognitive self-model, and evaluative judgment operator, and characterise unhappiness formally as an *identification error*—a logically unwarranted generalisation from a temporally bounded affective state to a global negative self-determination. Three canonical patterns of hypergeneralisation (temporal, scope, and identity) are defined and shown to correspond to established constructs in clinical psychology. A dynamical-systems embedding generates predictions about critical transitions into depressive states, with parameters mapped onto established psychometric instruments. The framework is situated within the ongoing debate between constructionist and basic-emotion accounts of affect and is shown to be consistent with a broad range of findings in affective neuroscience, clinical psychology, cross-cultural emotion research, and the sociology of subjective well-being. We argue that the thesis, if correct, carries significant implications for therapeutic practice, public-health policy, and philosophical anthropology. The central conclusion is that suffering is biologically obligate whereas unhappiness is cognitively facultative, and that the space between the two constitutes the domain of cognitive freedom.

Keywords: unhappiness; negative self-determination; cognitive appraisal; predictive processing; Stoic philosophy; dynamical systems; acceptance and commitment therapy; subjective well-being; meta-cognition

Contents

1 Introduction

The question "What makes people unhappy?" has occupied moral philosophy since antiquity [7, 93], developmental psychology since Freud [39], and empirical well-being research since the surveys of Bradburn [16]. Despite this long history, the literature still lacks a compact, formally stated principle that isolates the *sufficient and necessary cognitive act* through which unhappiness is generated.

The present paper proposes such a principle. We advance the thesis that *the act of negative self-determination is a principal source of unhappiness.* By "negative self-determination" we mean the reflexive cognitive operation in which a subject takes a first-order state—pain, loss, fatigue, frustration, fear—and converts it into a self-referential verdict: "I am unhappy," "This is intolerable," "I am defeated."

This thesis does *not* deny the reality of pain, grief, deprivation, or physiological suffering. It distinguishes, however, between *suffering* as a first-order experiential fact and *unhappiness* as a second-order interpretive act. The claim is that the latter arises only when the former is processed through a particular self-referential operator.

Such a distinction is not without precedent. The Stoic tradition, especially Epictetus and Marcus Aurelius, held that it is not events but our judgments about events that disturb us [77, Epictetus, Aurelius]. Buddhist psychology similarly separates primary sensation (*vedanā*) from the reactive craving and aversion (*taṇhā*) that produce sustained suffering (*dukkha*) [5, 44]. In contemporary cognitive psychology, the cognitive appraisal theories of Lazarus and Folkman [73, 74], Scherer [104, 105], and Smith and Ellsworth [112] establish that emotional experience depends critically on evaluative judgments rather than on raw stimulus properties.

What the present paper adds to these traditions is a *formal apparatus*—a small set of definitions, operators, and propositions stated with enough precision to generate falsifiable predictions—together with a systematic integration of evidence from predictive processing neuroscience, acceptance and commitment therapy, cross-cultural well-being research, and dynamical-systems theory.

The remainder of the paper is organised as follows. Section 2 provides a substantial literature review spanning philosophy, psychology, neuroscience, and sociology. Section 3 introduces the formal framework. Section 4 states and proves the main propositions. Section 5 embeds the framework in a dynamical-systems model. Section 6 discusses empirical predictions and boundary conditions. Section 7 explores implications for therapy, policy, and philosophical anthropology. Section 8 situates the paper within a broader research programme. Section 9 concludes.

2 Literature Review

2.1 Philosophical Antecedents

The idea that unhappiness resides not in external circumstances but in inner judgment has deep roots. Among the Stoics, Epictetus famously stated that "Men are disturbed not by things, but by the views which they take of things" [Epictetus]. Marcus Aurelius repeatedly returned to the thought that the capacity to remove a complaint removes the harm itself [Aurelius]. Seneca, in *De Tranquillitate Animi*, argued that the mind's restlessness is self-inflicted through comparison and craving [Seneca]. Comprehensive accounts of Stoic emotion theory can be found in Long [77], Nussbaum [93], and Sorabji [115].

The Stoics, however, retained a positive account of *eudaimonia* grounded in virtue: happiness required the active exercise of reason in conformity with nature. Our framework departs from Stoicism in a significant respect. We do *not* require a positive content of happiness; we require only the *absence* of the negative self-determination operator. This is closer to certain strands of Epicureanism, in which the highest good is *ataraxia*—the absence of disturbance [124, Epicurus].

Buddhist psychology offers a parallel analysis. The *Satipaṭṭhāna Sutta* teaches observation of sensations without identification: one notes "there is pain" rather than "I am in pain" [5]. The transition from bare awareness to self-referential clinging (*upādāna*) is identified as the mechanism that perpetuates suffering in the *pratītyasamutpāda* (dependent origination) framework [44, 49].

In modern Western philosophy, Sartre's analysis of "bad faith" (*mauvaise foi*) describes a self-deceptive fixation of identity that parallels the negative self-determination we formalise here [103]. Heidegger's notion of *Befindlichkeit* (attunement) suggests that mood is not a reaction to discrete events but a background mode of being-in-the-world that precedes propositional judgment [52]. Our framework complements this by proposing that the *transition* from pre-reflective attunement to reflective unhappiness requires an explicit self-determining act.

2.2 Cognitive Appraisal Theory

The formal study of cognitive mediation in emotion began with Arnold [8] and was elaborated by Lazarus in a series of landmark works [72–74]. The central thesis of appraisal theory is that an emotion is not elicited by an event per se but by the subject's evaluation of the event's significance for personal well-being. Lazarus distinguished *primary appraisal* (Is this event relevant to my goals?) from *secondary appraisal* (Can I cope?). The combination of high relevance and low coping capacity yields distress.

Smith and Ellsworth [112] identified six appraisal dimensions—pleasantness, anticipated effort, certainty, attentional activity, self-other responsibility, and situational control—that differentially predict discrete emotions. Scherer's component process model [104, 105] formalises appraisal as a sequential evaluation along stimulus evaluation checks (novelty, intrinsic pleasantness, goal relevance, coping potential, norm compatibility).

Within this tradition, our contribution is to isolate a *particular appraisal step*—the moment at which the subject identifies itself as "unhappy" or "defeated"—and to argue that this step is both necessary and sufficient for the emergence of unhappiness as a stable self-attributed state. Appraisal theory already implies that emotion depends on interpretation; we sharpen this to the claim that *unhappiness specifically* depends on a reflexive, identity-level verdict.

Empirical support for the appraisal framework is extensive. Roseman and colleagues demonstrated that manipulating appraisal components changes the reported emotion even when the eliciting situation is held constant [100]. Siemer, Mauss, and Gross [111] showed that appraisal patterns predict emotional responses above and beyond situation features. Moors and colleagues [86] provided a comprehensive review establishing appraisal theory as the dominant framework in emotion science.

2.3 Predictive Processing and Active Inference

The predictive processing framework, developed by Friston [40], Clark [26, 27], and Hohwy [56], models the brain as a prediction machine that continuously generates top-down expectations and compares them with bottom-up sensory signals. Discrepancies—prediction errors—drive either perceptual updating (changing the model) or active inference (changing the world to match the model).

Within this framework, affective states can be understood as interoceptive prediction errors: the brain predicts a certain bodily state, the actual state diverges, and the discrepancy is registered as an emotion [14, 109]. Barrett's theory of constructed emotion [14] proposes that discrete emotion categories are not natural kinds but conceptual acts that the brain uses to interpret interoceptive signals.

Our framework maps naturally onto predictive processing. The first-order state S_1 corresponds to an interoceptive prediction error. The negative self-determination operator $\mathcal{I}^-$ corresponds to the conceptual act by which this prediction error is interpreted as evidence that "I am unhappy." Crucially, the same prediction error could be interpreted differently—as challenge, as novelty, as temporary discomfort—without triggering the unhappiness label.

Seth's work on interoceptive inference [109, 110] is particularly relevant: he argues that emotional experience depends on how the brain models the causes of interoceptive signals. If the model attributes the signal to a transient, manageable cause, no unhappiness results. If the model attributes it to an enduring, identity-threatening cause, the conditions for negative self-determination are met.

2.4 Acceptance and Commitment Therapy

Acceptance and commitment therapy (ACT), developed by Hayes, Strosahl, and Wilson [50, 51], provides perhaps the closest clinical parallel to our framework. ACT identifies *cognitive fusion*—the process by which thoughts are taken to be literal truths about the self—as a core mechanism of psychopathology. The complementary process, *defusion*, involves stepping back from thoughts and observing them as mental events rather than as facts about identity.

The six core processes of ACT—acceptance, defusion, present-moment awareness, self-as-context, values, and committed action—can all be read as techniques for interrupting the negative self-determination operator. "Acceptance" is the refusal to appraise a first-order state as intolerable. "Defusion" is the refusal to fuse with the judgment "I am unhappy." "Self-as-context" replaces the content-rich self ("I am a person to whom bad things happen") with an observing self that does not identify with any particular state.

The empirical base for ACT is now substantial. Meta-analyses by A-Tjak and colleagues [10] and by Gloster and colleagues [45] report moderate to large effect sizes across anxiety, depression, chronic pain, and substance use, with the more recent review confirming consistent benefits across clinical and non-clinical populations. The mediational analyses consistently show that changes in psychological flexibility—the construct most closely linked to the absence of negative self-determination—mediate clinical outcomes [60, 76].

2.5 Subjective Well-Being and the Adaptation Paradox

One of the most robust findings in the science of subjective well-being (SWB) is the *adaptation effect*: individuals who experience major life changes—both positive and negative—tend to return toward their prior level of life satisfaction over time [28, 80, 81]. The landmark study by Brickman, Coates, and Janoff-Bulman [18] reported that lottery winners were not significantly happier than controls and that accident victims were not as unhappy as might be expected.

From the standpoint of our framework, adaptation is precisely what is predicted. If unhappiness depends on the act of negative self-determination rather than on objective circumstances, then changing circumstances without changing the operator leaves the long-run equilibrium unchanged. Conversely, if circumstances worsen but the operator is not engaged, unhappiness does not emerge.

Subsequent work has refined the adaptation picture. Lucas [79, 80] showed that adaptation to some events (e.g., widowhood, disability) is incomplete, while Diener, Lucas, and Scollon [30] argued for a "revised set-point theory" in which set points can change.

These refinements are compatible with our framework: incomplete adaptation may reflect situations in which the negative self-determination operator is persistently reactivated by environmental cues.

Cross-cultural studies further support the mediating role of interpretation. The Easterlin paradox—the observation that rising national income does not proportionally increase national happiness [32, 33]—suggests that absolute material conditions are less important than the cognitive framing of those conditions. Relative deprivation theory [102, 113] and social comparison theory [22, 38] both imply that unhappiness is generated by evaluative comparison, not by objective status.

2.6 Neuroscience of Self-Referential Processing

The default mode network (DMN), particularly the medial prefrontal cortex (mPFC) and posterior cingulate cortex (PCC), has been consistently implicated in self-referential processing [6, 21, 97]. Increased DMN activity during rest is associated with mind-wandering, rumination, and self-focused thought [61, 92].

Crucially, the relationship between DMN activity and unhappiness is mediated by the *evaluative quality* of self-referential thought. Killingsworth and Gilbert [61] showed that mind-wandering per se predicts lower happiness, but subsequent work distinguished between positive, neutral, and negative mind-wandering [6, 101]. It is specifically *negative evaluative self-referential thought*—rumination in the sense of Nolen-Hoeksema [90, 92]—that predicts depressive affect.

Meditation-based interventions that reduce self-referential processing, such as mindfulness-based stress reduction (MBSR) and mindfulness-based cognitive therapy (MBCT), have been shown to reduce DMN activity during self-referential tasks [17, 37] and to reduce depressive relapse [71, 117]. These findings are consistent with the claim that interrupting the negative self-determination operator—the evaluative, identity-laden processing of first-order states—reduces unhappiness.

2.7 Rumination and Depression

The response styles theory of Nolen-Hoeksema [90, 92] identifies *rumination*—repetitive, self-focused, evaluative thinking about one's depressive symptoms—as a key factor in the onset and maintenance of depression. Rumination is essentially a sustained, involuntary form of negative self-determination: the subject repeatedly tells itself "I am depressed," "Something is wrong with me," "This will not get better."

Empirical work has confirmed that rumination prospectively predicts depression [1, 91], that it amplifies the effect of negative life events on mood [99], and that interventions targeting rumination (such as rumination-focused cognitive-behavioural therapy) are effective in reducing depressive symptoms [125, 126].

Treynor, Gonzalez, and Nolen-Hoeksema [119] distinguished between *brooding*—passive comparison of one's current situation with an unachieved standard—and *reflection*—purposeful turning inward to solve problems. Only brooding, which maps directly onto our negative self-determination operator, predicted depressive symptoms. Reflection, which lacks the self-defeating verdict, did not.

2.8 Cross-Cultural Perspectives on Suffering and Selfhood

The proposed framework raises the question of whether the negative self-determination operator operates identically across cultures. Research on cultural variations in self-construal [82] distinguishes between independent self-construal (predominant in Western, educated, industrialised, rich, and democratic societies) and interdependent self-construal (more common in East Asian, South Asian, and many other cultural contexts).

The implications for our framework are nuanced. In cultures with interdependent self-construal, the "self" that is determined may be constituted partly by relational and communal identities [82]. The negative self-determination operator might therefore be triggered not only by threats to the individual ego but by threats to the relational self. Kitayama and colleagues [68, 69] showed that Japanese participants reported more "socially engaging" emotions (e.g., shame, guilt) as predictors of well-being, while American participants reported more "socially disengaging" emotions (e.g., pride, anger).

Importantly, however, the *structure* of the operator remains the same across cultures: unhappiness arises when a first-order state is processed through a self-referential verdict of defeat. The *content* of "self" may differ, but the *formal operation* is invariant. This is consistent with findings from the World Values Survey and Gallup World Poll suggesting that the predictors of life satisfaction are remarkably consistent across nations once basic needs are met [31, 54].

2.9 The Basic Emotions Challenge and the Constructionist–Nativist Debate

The framework advanced in this paper is most naturally aligned with constructionist accounts of emotion [14], which hold that discrete emotional categories are conceptual acts imposed on undifferentiated core affect rather than natural kinds with dedicated neural circuits. It is essential, however, to acknowledge that constructionism remains contested and that the alternative—the basic emotions tradition—poses a direct challenge to our thesis that must be addressed rather than bypassed.

The basic emotions programme, initiated by Ekman [34] and developed by Izard [57, 58] and Panksepp [94, 95], holds that certain emotional states (typically anger, fear, sadness, disgust, surprise, and happiness) are innate, universal, and subserved by distinct neural circuits. Panksepp's affective neuroscience programme identified seven primary-process emotional systems—SEEKING, RAGE, FEAR, LUST, CARE, PANIC/GRIEF, and PLAY—each with a characteristic subcortical substrate [94]. The PANIC/GRIEF system, in particular, generates states of distress, desolation, and social pain that many subjects and observers would describe as "unhappiness" in the ordinary-language sense, and these states appear to operate largely independently of the kind of propositional, self-referential cognition that our operator $\mathcal{I}^-$ requires.

This challenge must be met on two levels. At the terminological level, we note that our framework defines unhappiness as a *specific* meta-cognitive state—the self-referential verdict of defeat—rather than as a synonym for any negative affect. The distress generated by the PANIC/GRIEF system is, in our framework, a first-order affective state A (Definition 3.10): it is genuine suffering, biologically produced and morally significant, but it is not yet unhappiness in the technical sense we employ. A bereaved individual whose GRIEF system is active experiences profound anguish; whether that anguish becomes *unhappiness*—a settled conviction that "I am an unhappy person," "My life is ruined"—depends on whether the evaluative judgment operator J is activated.

At the substantive level, we acknowledge that this terminological distinction may seem like an evasion. The ordinary-language concept of "unhappiness" does not neatly separate first-order distress from second-order self-evaluation. Our theory makes a stipulative move: it reserves the term for the cognitively mediated case and explicitly places non-cognitive negative affect outside its scope. This is not a defect but a deliberate modelling choice. Just as thermodynamics does not claim to describe every form of energy but restricts itself to thermal phenomena, so our framework restricts itself to the cognitively mediated pathway through which adverse experience becomes a self-attributed state of unhappiness.

The residual—the non-cognitive contribution to negative affect—is formally captured by the term ε introduced in Definition 3.6 below. We do not deny its existence; we argue that the operator-mediated pathway is the principal one in adult human experience and the one most amenable to intervention.

The debate between constructionism and basic-emotion theory remains unresolved. Recent integrative proposals [3] suggest that the truth may involve both innate affective primitives and constructive conceptual processes. Our framework is compatible with such hybrid accounts: the basic-emotion systems generate the first-order input A, while the constructive process corresponds to the evaluative judgment J. What our theory adds is the claim that it is the second step, not the first, that transforms suffering into the self-attributed state of unhappiness.

2.10 Emotion Regulation Theory and Metacognitive Therapy

Gross's process model of emotion regulation [128] identifies five families of regulatory strategies ordered by their point of intervention in the emotion-generative process: situation selection, situation modification, attentional deployment, cognitive change, and response modulation. The present framework maps naturally onto this taxonomy. Situation selection and modification target the input function $f(t)$—they aim to reduce the first-order suffering signal before it arises. Attentional deployment targets the coupling parameter γ by redirecting attention away from stimuli that activate the operator. Cognitive change—the strategy most characteristic of CBT—targets both the rumination parameter δ and the content of the evaluative verdict $\mathcal{V}^-$, corresponding to what Gross terms "reappraisal." Response modulation targets λ, the translation of operator–suffering co-occurrence into experienced unhappiness.

This mapping is not merely descriptive. The dynamical model yields a quantitative prediction that Gross's framework alone does not: because δ enters the equations multiplicatively and governs the critical transition (Proposition 5.1), interventions targeting cognitive change (i.e., reducing δ) should show disproportionately large effects near the bifurcation threshold, while the same intervention should show smaller effects in individuals far from criticality. This predicts a treatment-by-severity interaction that is testable in existing datasets.

Wells's metacognitive therapy (MCT; 127) provides an even more direct connection. MCT distinguishes between *object-level* cognitive content ("I am a failure") and *meta-level* beliefs about cognition ("worrying helps me prepare"; "I cannot control my thoughts"). In our framework, object-level content corresponds to the verdict $\mathcal{V}^-$, while meta-level beliefs correspond to parameters of the operator itself—specifically, δ (positive meta-beliefs about rumination increase the self-reinforcing loop) and β (negative meta-beliefs about uncontrollability decrease the decay rate of the operator). The memory term $m(t)$ in Equation (3) formalises what Wells calls the "Cognitive Attentional Syndrome" (CAS): a self-perpetuating configuration of worry, rumination, threat monitoring, and unhelpful coping that sustains distress beyond the original trigger. MCT's clinical strategy—interrupting the CAS by modifying meta-beliefs rather than object-level content—is precisely an intervention on the operator's parameters rather than on its outputs, a distinction the present formalism makes explicit.

3 Formal Framework

We now introduce the formal apparatus of the theory. The notation is deliberately kept at the level of elementary set theory and operator algebra so that readers from psychology, sociology, and biology can follow the reasoning without specialised training in dynamical systems. Each definition and proposition is accompanied by a detailed verbal exposition that clarifies the intuitive meaning and the empirical motivation.

Scope of the formalism. The formal model developed in this section describes the *cognitively mediated pathway* through which adverse experience is transformed into self-attributed unhappiness. As discussed in Section 2.9, we acknowledge that non-cognitive processes (e.g., neurochemical dysregulation, subcortical affective circuits) can independently contribute to negative affective states. These contributions are captured by a residual term introduced in Definition 3.6. The formalism is designed to model the component of unhappiness that is most prevalent in everyday adult human experience and most amenable to psychological intervention.

3.1 Primitive Concepts

We begin by specifying the basic objects of the theory.

Definition 3.1 (Event Space). *Let $\mathcal{E}$ denote the set of all events that may befall a subject. An event $E \in \mathcal{E}$ is any occurrence in the external or internal environment of the subject, including physical injury, social loss, environmental change, interoceptive signals, and cognitive intrusions.*

The event space is intentionally broad. It includes not only external occurrences—such as the loss of a job, the death of a loved one, or exposure to a natural disaster—but also internal events such as the onset of a headache, a surge of anxiety, or an intrusive thought. The crucial point is that events, as we define them here, are ontologically prior to any evaluative act by the subject.

Definition 3.2 (First-Order State Space). *Let $\mathcal{S}$ denote the set of first-order experiential states available to the subject. A first-order state $S \in \mathcal{S}$ is any sensory, affective, or physiological condition directly produced by an event or by the body's response to an event. Examples include pain, fatigue, hunger, fear arousal, grief pangs, and muscular tension.*

First-order states are the immediate, pre-reflective consequences of events as experienced by the subject. They are "first-order" in the sense that they do not yet involve a judgment about the self. Pain is a first-order state: it is experienced directly and does not require the subject to form a belief about its own identity. Similarly, the physiological arousal associated with fear—elevated heart rate, cortisol release, startle response—is a first-order state that occurs automatically.

Definition 3.3 (Event–State Mapping). *Let $\varphi : \mathcal{E} \to \mathcal{S}$ be the mapping from events to first-order states. For an event E, the associated first-order state is $S = \varphi(E)$. This mapping is generally many-to-one: multiple events may produce the same first-order state.*

This mapping captures the empirical fact that different events can produce similar experiential states. A job loss, a romantic rejection, and a medical diagnosis may all produce the first-order state of anxious distress. Conversely, the same event (e.g., rainfall) may produce different first-order states depending on context (relief during drought, distress during a flood). The mapping φ is therefore context-dependent, but for present purposes, we treat it as given.

3.2 The Negative Self-Determination Operator

We now introduce the central construct of the theory.

Definition 3.4 (Negative Self-Determination Operator). *Let $\mathcal{I}^- : \mathcal{S} \to \{0, 1\}$ be a binary operator called the* negative self-determination operator. *For a first-order state $S \in \mathcal{S}$,*

$$\mathcal{I}^-(S) = \begin{cases} 1 & \textit{if the subject forms the reflexive judgment "I am unhappy/defeated,"} \\ 0 & \textit{otherwise.} \end{cases}$$

This operator captures the precise cognitive act that, according to our theory, is responsible for the emergence of unhappiness. When the operator takes the value 1, the subject has moved from experiencing a first-order state (e.g., pain, loss, frustration) to forming a self-referential judgment that identifies itself as "unhappy," "broken," "a failure," or some equivalent verdict of personal defeat.

The judgment need not be linguistically explicit. It may occur as an implicit felt sense of "being unhappy" or as a non-verbal identification with a defeated self-image. What matters is the *structural* feature: the subject takes itself as an object of evaluation and assigns a negative verdict. This is what distinguishes simple suffering from unhappiness as a self-attributed state.

Definition 3.5 (Unhappiness). *Given an event $E \in \mathcal{E}$ with associated first-order state $S = \varphi(E)$, the unhappiness U of the subject is defined as:*

$$U(E) = \mathcal{I}^-\big(\varphi(E)\big).$$

That is, unhappiness arises if and only if the negative self-determination operator is activated on the first-order state produced by the event.

This definition makes the theory's core claim formally explicit. Unhappiness is a *composite* phenomenon: it requires both a first-order state S (produced by some event E) and the activation of the operator $\mathcal{I}^-$. Neither component alone is sufficient. An event that produces no first-order distress (e.g., a minor inconvenience) typically does not activate the operator. A first-order state that is experienced without self-referential evaluation (e.g., pain during athletic exertion that is framed as "effort" rather than "suffering") does not produce unhappiness.

To accommodate non-cognitive contributions to negative affect (see Section 2.9), we introduce the full model:

Definition 3.6 (Full Unhappiness Model)**.** *The total negative affective state experienced by a subject in response to event E is:*

$$U_{total}(E) = \underbrace{\mathcal{I}^{-}\big(\varphi(E)\big)}_{U_{cognitive}} + \underbrace{\varepsilon(E)}_{residual}$$

where $\varepsilon(E) \in [0,1]$ captures non-cognitive contributions to negative affect—neurochemical dysregulation (as in endogenous depression), subcortical affective reactions (as in Panksepp's GRIEF/PANIC system), pharmacological effects, and neuropathological processes. The present theory models $U_{cognitive}$; the residual ε is acknowledged but lies outside the theory's explanatory scope.

The claim of this paper is that, in the vast majority of everyday human experience, $U_{\text{cognitive}} \gg \varepsilon$—that is, the operator-mediated pathway dominates the residual. The residual term becomes significant primarily in clinical populations with neurobiological pathology (e.g., Parkinson's disease anhedonia, pharmacologically induced dysphoria, severe traumatic brain injury). Importantly, even in these cases the operator-mediated component often remains active and amenable to intervention [25, 51].

3.3 Generalisation to Continuous Intensity

The binary formulation above is a useful idealisation, but real-world unhappiness admits of degrees. We therefore introduce a continuous generalisation.

Definition 3.7 (Continuous Negative Self-Determination)**.** *Let $\mathcal{I}_c^{-} : \mathcal{S} \to [0,1]$ be the continuous form of the negative self-determination operator, where $\mathcal{I}_c^{-}(S) = 0$ indicates no self-referential negative verdict and $\mathcal{I}_c^{-}(S) = 1$ indicates maximal identification with defeat. The continuous unhappiness function is then:*

$$U_c(E) = \mathcal{I}_c^{-}\big(\varphi(E)\big) \in [0,1].$$

This continuous formulation accommodates the everyday observation that people are not simply "happy" or "unhappy" in a binary sense but experience varying degrees of unhappiness. A mild headache accompanied by a fleeting thought of "this is annoying" might correspond to $\mathcal{I}_c^{-} \approx 0.1$, whereas a severe illness accompanied by sustained self-identification as "broken" and "hopeless" might correspond to $\mathcal{I}_c^{-} \approx 0.9$.

3.4 The Suffering–Unhappiness Distinction

We now formalise the key distinction between suffering and unhappiness.

Definition 3.8 (Suffering)**.** *The* suffering *associated with an event E is the first-order state $S = \varphi(E)$ itself. Suffering is a property of the experiential stream and does not require any meta-cognitive act.*

Definition 3.9 (The Suffering–Unhappiness Gap)**.** *The* suffering–unhappiness gap *for an event E is defined as the case in which $\varphi(E) \neq \mathbf{0}$ (i.e., there is genuine first-order distress) but $\mathcal{I}^-(\varphi(E)) = 0$ (i.e., the negative self-determination operator is not activated). In this case, the subject suffers but is not unhappy.*

This gap is central to the theory and is well-documented empirically. Chronic pain patients who practice acceptance report significant pain but reduced distress and improved quality of life [84, 122]. Experienced meditators report pain intensity comparable to controls but dramatically reduced unpleasantness [48, 130]. Terminal patients who achieve acceptance frequently report equanimity or even gratitude despite severe physical decline [25, 70]. In each case, the first-order state is present, but the negative self-determination operator is not engaged, and the result is suffering without unhappiness.

3.5 The Three-Component Refined Model

The binary operator $\mathcal{I}^-$ introduced above captures the essential mechanism, but a finer-grained analysis reveals that unhappiness is generated by the interaction of three distinct components rather than by a single monolithic act. These components correspond to three levels of processing that are well-established in affective science and that map onto distinguishable neural substrates.

Definition 3.10 (Primary Affective Reaction)**.** *Let $A \in \mathcal{A}$ denote the* primary affective reaction*—the automatic, biologically determined emotional response elicited by an event. This includes fear, pain, startle, grief pangs, and other rapid-onset states mediated primarily by subcortical circuits (amygdala, periaqueductal grey, hypothalamus). The reaction A is reflexive and does not require deliberation.*

Primary affective reactions are the organism's first line of response. They evolved to serve survival functions: fear prompts flight, pain prompts withdrawal, grief signals the loss of a bonded other [75, 94]. These reactions arise automatically and cannot be prevented by an act of will. A sudden loud noise produces startle; the death of a child produces grief pangs. No theory of unhappiness should deny or trivialise these biological facts.

Definition 3.11 (Cognitive Model of Self)**.** *Let $M \in \mathcal{M}$ denote the subject's* cognitive model of self*—the structured representation of "who I am," including beliefs about personal identity, narrative self-understanding, life trajectory, social roles, and core values. The model M is continuously maintained and updated, and it provides the background against which affective reactions are interpreted.*

The cognitive model of self is the accumulated product of a lifetime of experience, socialisation, and reflection. It contains propositions such as "I am a competent professional," "I am a good parent," "My life has meaning." It also contains implicit standards and expectations—the predictive models, in the language of predictive processing, against which incoming signals are evaluated [27, 40].

Definition 3.12 (Evaluative Judgment Operator)**.** *Let $J : \mathcal{A} \times \mathcal{M} \to [0, 1]$ be the* evaluative judgment operator *that takes as input a primary affective reaction A and the cognitive model of self M, and produces an output in $[0, 1]$ representing the degree to which the subject concludes "this state means that I am unhappy." The unhappiness generated by the three-component model is then:*

$$U = J(A, M).$$

The evaluative judgment operator J is the refined version of $\mathcal{I}^-$. It makes explicit that the transition from suffering to unhappiness depends on two inputs simultaneously: the affective reaction A and the cognitive model M. The same affective reaction, interpreted against different cognitive models, yields different degrees of unhappiness. A professional setback produces high J in a subject whose cognitive model centres on career achievement and low J in a subject whose model does not.

The relationship between the original binary operator and the three-component model is given by $\mathcal{I}^-(S) = \mathbf{1}[J(A, M) > \theta]$, where θ is a threshold and $\mathbf{1}[\cdot]$ is the indicator function. The three-component model is thus a refinement, not a replacement, of the original framework.

3.6 Unhappiness as Identification Error

With the three-component model in place, we can now give a precise characterisation of the cognitive operation that generates unhappiness.

Definition 3.13 (Identification Error)**.** *An* identification error *occurs when the evaluative judgment operator J maps a temporally bounded, state-level affective reaction onto a global, identity-level conclusion about the self. Formally, an identification error is a mapping of the form:*

$$J : A_{local}(t) \longmapsto M^-_{global}$$

where $A_{local}(t)$ is a time-indexed, situation-specific affective state and M^-_{global} is a negative revision of the global self-model.

The identification error is a category mistake with a specific logical structure. A transient state ("I feel bad right now") is treated as evidence for a permanent identity ("I am an unhappy person"). A local failure ("This project did not succeed") is generalised to a global verdict ("I am a failure"). A situational difficulty ("This is hard") is inflated to an existential conclusion ("This is unbearable; my life is unbearable").

This structure can be further decomposed into three characteristic patterns of hypergeneralisation.

Definition 3.14 (Hypergeneralisation Patterns)**.** *Three canonical forms of identification error can be distinguished, each involving a logically unwarranted leap from a bounded observation to an unbounded conclusion:*

(i) Temporal hypergeneralisation: *"I feel bad now" $\longrightarrow$ "I will always feel bad."*
The temporal boundary of the affective state is removed. A transient condition is projected onto the indefinite future.

(ii) Scope hypergeneralisation: *"This situation is bad" $\longrightarrow$ "My whole life is bad."*
The situational boundary of the affective state is removed. A domain-specific problem is projected onto the totality of the subject's existence.

(iii) Identity hypergeneralisation: *"I failed at this" $\longrightarrow$ "I am a failure."*
The action boundary is removed. A discrete behaviour or outcome is projected onto the subject's core identity.

These three patterns are well-documented in clinical psychology. Beck's cognitive triad [15] identified negative views of the self, the world, and the future as hallmarks of depressive cognition—a structure that maps directly onto the three hypergeneralisation patterns. Burns [23] catalogued "cognitive distortions" including overgeneralisation, labelling, and fortune-telling, all of which are instances of the identification error defined here. Abramson, Seligman, and Teasdale's reformulated learned helplessness model [2] identified internal, stable, and global attributional style as a vulnerability factor for depression—a tripartite structure that corresponds precisely to identity, temporal, and scope hypergeneralisation, respectively.

3.7 The Predictive Processing Interpretation of Identification Error

The identification error acquires additional explanatory force when situated within the predictive processing framework. In predictive processing, the brain continuously generates top-down predictions about incoming sensory (including interoceptive) signals and updates its generative model when prediction errors are detected [27, 40, 109].

Under normal conditions, a prediction error triggers a *local* model update: the specific prediction that failed is revised, and the system returns to equilibrium. The identification error, however, corresponds to a *catastrophic hypergeneralising interpretation of prediction error.* Instead of revising a local component of the model ("my prediction about this situation was wrong"), the system revises the global self-model ("I am the kind of person to whom bad things happen"; "I am fundamentally inadequate").

This distinction can be formalised as follows.

Definition 3.15 (Local vs. Global Model Update)**.** *Let $\varepsilon(t)$ denote a prediction error at time t. A* local update *revises only the prediction parameters directly associated with the error:*

$$M_{t+1} = M_t + \eta_{local} \nabla_{M_{local}} \varepsilon(t)$$

where η_{local} is a local learning rate and $\nabla_{M_{local}}$ is the gradient with respect to local model parameters. A global (catastrophic) update *revises the self-model at the identity level:*

$$M_{t+1} = M_t + \eta_{global} \nabla_{M_{self}} \varepsilon(t)$$

where η_{global} is the global learning rate and $\nabla_{M_{self}}$ is the gradient with respect to core self-model parameters.

Under this analysis, the negative self-determination operator is activated precisely when $\eta_{\text{global}} \gg \eta_{\text{local}}$—that is, when the system responds to a prediction error by revising its deepest self-representations rather than the local predictions that actually failed. This is computationally inefficient (it overfits the identity model to transient data), experientially devastating (it destabilises the subject's sense of self), and logically unwarranted (a single prediction error does not provide sufficient evidence for a global model revision).

The conditions under which the system defaults to global rather than local updating are of considerable clinical interest. Early adverse experiences may establish prior expectations that favour global negative self-revision [53]. Insecure attachment may reduce the precision-weighting of positive self-related predictions, making the self-model more vulnerable to destabilisation by negative evidence [85]. Chronic stress may deplete the computational resources required for fine-grained local updating, causing the system to fall back on cruder, more global revision strategies [9].

3.8 The Error Criterion and the Principle of Facultative Unhappiness

The characterisation of negative self-determination as a "cognitive error" requires a criterion of correctness. Without such a criterion, the label "error" would be merely an alternative interpretation, no more privileged than the one it seeks to displace.

We propose the following criterion.

Definition 3.16 (Error Criterion for Self-Determination)**.** *A self-determination act* $J(A, M)$ *is an* error *if and only if the scope of the conclusion exceeds the scope of the evidence. Specifically,* $J(A, M)$ *constitutes an error when a temporally bounded, situation-specific affective state* $A_{local}(t)$ *is used as the sole basis for a global, temporally unbounded, identity-level revision* M^{-}_{global}.

This criterion is not arbitrary. It rests on the elementary logical principle that a universal conclusion cannot be validly derived from a particular premise. "I feel pain now" does not entail "I am an unhappy person," just as "It rained today" does not entail "It always rains." The error is one of *scope*: the conclusion is broader than the evidence warrants.

With this criterion in place, we can state the principle that emerges from the analysis:

Proposition 3.1 (The Facultative Nature of Unhappiness)**.** *Suffering—the primary affective reaction to adverse events—is biologically determined and, to a substantial degree, involuntary. Unhappiness—the global negative self-determination that results from the identification error—is facultative: it depends on the activation of a cognitive operator that is, in principle, interruptible. Formally:*

$$\textit{Suffering is obligate:} \quad A = \varphi_A(E), \quad \textit{determined by } E.$$

$$\textit{Unhappiness is facultative:} \quad U = J(A, M), \quad \textit{contingent on } J.$$

The space between the obligate and the facultative is the space of cognitive freedom.

This proposition does not deny the reality of suffering or the severity of adverse events. It does not claim that one can choose to feel no pain or to suppress grief by an act of will. It claims, rather, that the *second-order transition*—from experiencing a negative state to identifying oneself as a globally unhappy person—is not a necessary consequence of the first-order state. It is a cognitive act that can, in principle, fail to occur; and when it does occur, it constitutes an error by the criterion defined above.

The principle "suffering is obligate; unhappiness is facultative" captures the theory's central implication. The inevitability of suffering is not in question. What is in question—and what the theory answers—is whether the move from suffering to unhappiness is equally inevitable. The answer is no. And it is in this gap between the obligate and the facultative that the possibility of a life free from unhappiness, though not from suffering, resides.

4 Main Propositions

We now state and establish the principal results of the theory.

Proposition 4.1 (Dominance of the Cognitive Pathway)**.** *For any event E producing a first-order state $\varphi(E)$ in a neurologically typical adult, the cognitive component of unhappiness dominates the residual:*

$$U_{cognitive}(E) \geq (1 - \varepsilon_0)\, U_{total}(E)$$

for some $\varepsilon_0 \ll 1$ that depends on the subject's neurobiological integrity. That is, the negative self-determination operator accounts for the overwhelming share of experienced unhappiness.

This proposition is not a definitional consequence but an empirical claim that generates testable predictions. It asserts that in healthy adult populations, the variance in unhappiness is explained predominantly by cognitive-appraisal processes rather than by raw neurochemical or subcortical contributions. The proposition would be falsified if, for example, a pharmacological agent that selectively blocks self-referential cognition (without affecting first-order affect) failed to reduce self-reported unhappiness, or if neurochemical manipulations that bypass cognitive appraisal reliably produced sustained self-attributed unhappiness rather than merely transient negative mood.

Addressing the circularity concern. A careful reader may object that if "unhappiness" is *defined* as a cognitively mediated state, then the claim that the cognitive operator dominates unhappiness is true by definition. This concern is legitimate and requires a clear response. The theory involves two logically distinct components.

The first component is a *conceptual proposal*: that the ordinary-language term "unhappiness" tracks a cognitive-evaluative state (the self-referential verdict) rather than raw negative affect, and that ordinary usage already implicitly distinguishes these. This is a philosophical claim about the structure of the concept, supported by the semantic evidence that people distinguish "being in pain" from "being unhappy."

The second component is an *empirical claim*: that the variance in global life-dissatisfaction self-reports (SWLS, inverse) in neurologically typical adults is explained predominantly by measures of self-referential cognitive processing (RRS-brooding, EQ decentring, FFMQ non-reactivity) rather than by measures of first-order affective intensity (McGill Pain sensory subscale, PANAS negative affect momentary ratings, cortisol AUC). This is a testable claim. Specifically, in a hierarchical regression predicting SWLS scores, the addition of self-referential processing measures after first-order affective measures should explain at least twice the incremental variance ($\Delta R^2_{\text{cognitive}} \geq 2 \times \Delta R^2_{\text{affective}}$). This prediction can be falsified; the conceptual proposal alone cannot guarantee this result, since it is logically possible that people form the self-referential verdict but that the verdict explains little variance in their global life assessments.

The empirical evidence reviewed in Section 2.4 and Section 2.7 supports this claim.

ACT interventions that promote defusion (i.e., disengagement of $\mathcal{I}^-$) reduce unhappiness even when objective circumstances are unchanged. Rumination research shows that the repetitive, self-referential rehearsal of negative verdicts (i.e., sustained activation of $\mathcal{I}^-$) is what maintains depression, not the triggering event itself.

Proposition 4.2 (Sufficiency of the Operator)**.** *The activation of the negative self-determination operator is sufficient for unhappiness, even in the absence of a proportionate first-order state. Formally, there exist cases in which* $\varphi(E) \approx \mathbf{0}$ *yet* $\mathcal{I}^-(S) = 1$ *and* $U(E) = 1$*.*

This proposition captures the phenomenon of unhappiness without proportionate cause—a condition familiar from clinical depression, generalised anxiety disorder, and existential crisis. A person may have adequate physical health, financial security, and social support, yet form the self-referential judgment "I am unhappy," "My life is meaningless," or "Something is fundamentally wrong with me." In such cases, the first-order state is mild or even neutral, but the operator is fully engaged.

We note that this proposition requires a slight extension of the framework. The operator $\mathcal{I}^-$ can be activated not only by current first-order states but also by *anticipated* or *remembered* states—a phenomenon extensively documented in the affective forecasting literature [47, 129]. People frequently become unhappy about projected future losses or recalled past injuries, even when their present first-order state is comfortable. The operator acts on the *represented* state, not only on the currently experienced one.

Proposition 4.3 (Non-Transitivity of Events to Unhappiness)**.** *The mapping from events to unhappiness is not transitive through severity. Formally, for events* $E_1, E_2 \in \mathcal{E}$*, even if* $\varphi(E_1)$ *is objectively more severe than* $\varphi(E_2)$*, it is possible that* $U(E_1) = 0$ *and* $U(E_2) = 1$*.*

This proposition formalises the well-known observation that people sometimes cope better with major adversities than with minor ones. A soldier may endure grievous injury without self-identifying as unhappy (because the injury is framed as honourable sacrifice), while the same soldier may be devastated by a minor social slight (because it activates a narrative of rejection and worthlessness). The disability paradox—the finding that many individuals with severe disabilities report high life satisfaction [4, 121]—is a large-scale instance of this non-transitivity.

The formal explanation is straightforward. The mapping from event severity to operator activation is not monotone. The operator is triggered not by the magnitude of the first-order state but by the *meaning* the subject assigns to it, which depends on personal history, cultural context, narrative identity, and available coping frameworks.

Proposition 4.4 (Compositional Structure of Negative Self-Determination)**.** *The negative self-determination operator can be decomposed into two sub-operations:*

$$\mathcal{I}^- = \mathcal{V}^- \circ \mathcal{R}$$

where $\mathcal{R} : \mathcal{S} \to \mathcal{S}_{self}$ *is the* reflexive attention operator *(directing attention to the self as object) and* $\mathcal{V}^- : \mathcal{S}_{self} \to \{0, 1\}$ *is the* negative verdict operator *(assigning a negative evaluation to the self-as-object).*

This decomposition has practical value for therapeutic intervention. It suggests two distinct points of interruption. First, one can disrupt $\mathcal{R}$—the reflexive turn toward self-evaluation—as in mindfulness practices that redirect attention to the sensory present [59, 107]. Second, one can disrupt $\mathcal{V}^-$—the negative evaluation—as in cognitive restructuring [15] or self-compassion practices [87, 88].

The decomposition also maps onto neuroscientific findings. The reflexive attention operator $\mathcal{R}$ is associated with activation of the default mode network [21, 97]. The negative verdict operator $\mathcal{V}^-$ is associated with activity in the subgenual anterior cingulate cortex and amygdala during self-critical processing [78]. Mindfulness interventions reduce DMN activation [17], while cognitive-behavioural interventions target the evaluative content, providing independent modulation of the two sub-operators.

5 Dynamical-Systems Embedding

We now embed the operator-theoretic framework in a dynamical-systems model to capture how unhappiness evolves over time. The purpose of this section is to show that the theory can generate temporal predictions about the onset, maintenance, and resolution of unhappiness.

5.1 State Variables

We model the system using three continuously evolving state variables.

The first variable, $s(t) \in [0, 1]$, represents the intensity of the first-order suffering state at time t. A value of $s = 0$ indicates no first-order distress, while $s = 1$ represents maximal distress.

The second variable, $a(t) \in [0, 1]$, represents the activation level of the negative self-determination operator at time t. A value of $a = 0$ means the operator is completely inactive (the subject is not forming any self-referential negative judgment), while $a = 1$ means the operator is fully engaged.

The third variable, $u(t) \in [0, 1]$, represents the intensity of unhappiness at time t.

5.2 Equations of Motion

The evolution of the system is governed by the following system of ordinary differential equations. All state variables are confined to $[0, 1]$ by the use of saturating nonlinearities, ensuring that the model is well-defined throughout the state space including near critical transitions.

$$\frac{ds}{dt} = -\alpha\, s + (1 - s)\, f(t) \tag{1}$$

This equation describes how the first-order suffering state changes over time. The term $-\alpha\, s$ represents the natural decay of distress: in the absence of ongoing stimulation, pain, fear, and other first-order states tend to diminish at a rate proportional to their current intensity. The parameter $\alpha > 0$ is the decay rate, reflecting the body's homeostatic mechanisms. The function $f(t) \geq 0$ represents the external input—the ongoing stream of adverse events. The factor $(1 - s)$ ensures that s cannot exceed 1: as $s \to 1$, the input is attenuated, modelling the physiological ceiling on distress intensity.

$$\frac{da}{dt} = -\beta\, a + \gamma\, (1 - a)\, \big[s + \kappa\, m\big]\, g(a) \tag{2}$$

This equation describes how the activation level of the negative self-determination operator evolves. The term $-\beta\, a$ represents the natural decay of self-referential processing. The excitatory term $\gamma\, [s + \kappa\, m]\, g(a)$ captures the input from two sources: the current first-order state s and a *representational memory term* $m \in [0, 1]$, weighted by $\kappa \geq 0$. The memory term m represents anticipated or recalled distress—the cognitive representation of past or future suffering that can activate the operator even when current first-order input is low. The factor $(1 - a)$ is the saturation nonlinearity that confines a to $[0, 1]$: as operator activation approaches its maximum, further excitation is attenuated.

The self-reinforcement function is modelled as a saturating form: $g(a) = 1 + \delta\, a/(1 + a)$, where $\delta \geq 0$ is the *rumination parameter*. This function increases monotonically with a but saturates at $1 + \delta/2$ as $a \to 1$, ensuring that the positive feedback loop does not produce a finite-time blow-up.

The memory term $m(t)$ evolves according to:

$$\frac{dm}{dt} = -\nu\, m + \eta\, a \tag{3}$$

where $\nu > 0$ is the decay rate of representational memory and $\eta > 0$ is the rate at which operator activation generates persistent cognitive representations. This equation captures the bidirectional relationship between rumination and memory: sustained negative self-determination $(a > 0)$ deposits representational residue (m), which in turn can re-excite the operator even after the original stressor has subsided.

$$\frac{du}{dt} = -\mu\, u + \lambda\, (1 - u)\, a\, (s + \kappa\, m) \tag{4}$$

This equation defines unhappiness as driven by the joint product of operator activation

a and the total distress signal $s + \kappa\, m$ (current suffering plus represented suffering). The factor $(1-u)$ provides saturation. Crucially, this formulation reconciles the model with Proposition 4.2: even when current suffering $s \approx 0$, if the memory term $m > 0$ (from past rumination) and the operator is active ($a > 0$), unhappiness u receives positive input. This accounts for existential depression, clinical rumination without proportionate current distress, and unhappiness driven by anticipated rather than experienced events.

5.3 Equilibrium and Bifurcation Analysis

The system admits a trivial equilibrium at $(s^*, a^*, m^*, u^*) = (0, 0, 0, 0)$, which represents the quiescent state in the absence of adverse events.

Under constant adverse input $f(t) = f_0 > 0$, the suffering state reaches a steady value:

$$s^* = \frac{f_0}{\alpha + f_0} \tag{5}$$

which lies in $(0, 1)$ for all $f_0 > 0$—the saturation nonlinearity ensures that s^* never exceeds 1 regardless of input intensity.

For the operator equilibrium, we set $da/dt = 0$ and $dm/dt = 0$. From Equation (3), $m^* = (\eta/\nu)\, a^*$. Substituting into the operator equation, we define the effective input $\sigma^* = s^* + \kappa\, (\eta/\nu)\, a^*$ and solve:

$$\beta\, a^* = \gamma\, (1 - a^*)\, \sigma^* \left(1 + \delta\, a^*/(1 + a^*)\right) \tag{6}$$

This is a nonlinear equation in a^* that can be analysed by defining the right-hand side as a function $F(a^*; \delta, f_0)$ and examining its intersections with the left-hand side $\beta\, a^*$. The bifurcation structure is as follows.

Subcritical regime ($\delta < \delta_{\text{crit}}$): The equation has a unique stable fixed point $a^* \in (0, 1)$. The Jacobian of the linearised system at this fixed point has all eigenvalues with negative real part, confirming local asymptotic stability. Unhappiness is finite and proportionate to stressor intensity.

Critical transition ($\delta = \delta_{\text{crit}}$): A saddle-node bifurcation occurs. The unique low-activation fixed point collides with an unstable fixed point and both disappear, leaving only a high-activation attractor near $a^* \approx 1$. This is a genuine dynamical bifurcation (not an algebraic pole), and it corresponds clinically to the abrupt onset of a depressive episode.

The critical value can be approximated for small $\kappa\, \eta/\nu$ as:

Proposition 5.1 (Rumination Criticality—Saturating Model)**.** *In the saturating model,* *the system undergoes a saddle-node bifurcation when δ satisfies approximately:*

$$\delta_{crit} \approx \frac{\beta}{\gamma\, s^*} - 1 + \frac{\kappa\,\eta}{\nu}\,\frac{\beta}{\gamma\,(s^*)^2}$$

where $s^* = f_0/(\alpha + f_0)$. For $\delta < \delta_{crit}$, the system has a stable low-activation equilibrium. For $\delta > \delta_{crit}$, the system transitions to a high-activation regime ($a^* \to 1$, $u^* \to 1$), representing pathological fixation of negative self-determination.

The Jacobian of the four-variable system (s, a, m, u) at any equilibrium can be computed as a 4×4 upper-triangular-block matrix (since s is decoupled from the other variables). The eigenvalue associated with s is $\lambda_1 = -(\alpha + f_0) < 0$, always stable. The remaining three eigenvalues govern the (a, m, u) subsystem and can be computed numerically for specific parameter values. The critical transition is characterised by one eigenvalue crossing zero from the negative half-plane—the hallmark of a saddle-node bifurcation.

This result generates a testable prediction: individuals with higher trait rumination (δ) should exhibit a lower threshold for the transition from manageable distress to clinical depression. Existing longitudinal data on rumination and depression onset are broadly consistent with this prediction [1, 91, 126].

5.4 Interpretation of Parameters

Each parameter in the model corresponds to a psychologically and biologically meaningful quantity.

The parameter α (decay rate of first-order suffering) is determined by neurobiological homeostatic mechanisms, including endogenous opioid release, cortisol regulation, and autonomic nervous system recovery. It may vary across individuals and can be modulated by pharmacological interventions.

The parameter β (decay rate of the operator) reflects the natural tendency for self-referential processing to wane. Individuals with strong dispositional mindfulness [19] or high psychological flexibility [60] may be expected to have higher β, meaning that their self-referential processing decays more quickly.

The parameter γ (coupling strength from suffering to operator) captures how readily first-order distress triggers self-referential evaluation. This may be influenced by attachment style [85], early adverse experiences [53], and cultural norms regarding emotional expression [83].

The parameter δ (rumination amplification) is the most clinically significant. It measures the degree to which the negative self-determination operator feeds back on itself. Cognitive-behavioural interventions [15, 126] and mindfulness-based interventions [107] can be understood as targeting this parameter specifically.

The parameters κ, η, and ν govern the representational memory subsystem. The weight κ captures the degree to which represented (anticipated or recalled) distress activates the operator relative to current distress. The rate η determines how quickly sustained rumination deposits memory traces, and ν determines how quickly those traces decay. High η/ν ratios characterise individuals who build persistent cognitive residue from rumination episodes—a pattern associated with trait rumination [92] and consistent with the metacognitive model of Wells [127], in which "meta-worry" (worry about worry) generates self-sustaining representational loops.

The parameter λ (coupling of operator and suffering to unhappiness) represents the efficiency with which the joint presence of distress and self-referential evaluation translates into experienced unhappiness. It may be modulated by factors such as emotional granularity [13, 120] and interoceptive sensitivity [43]. In the framework of Gross's process model of emotion regulation [128], the parameter λ is targeted by response-modulation strategies, while γ corresponds to attentional deployment and δ to cognitive change.

6 Empirical Predictions and Boundary Conditions

6.1 Testable Predictions and Operationalisation

The theory generates several predictions amenable to empirical investigation. To address concerns about falsifiability, we specify for each prediction the measurement instruments that operationalise the model's parameters and the quantitative form of the expected result.

A note on the status of the numerical estimates. The specific numerical thresholds stated below (e.g., "40–60% lower," "$\geq$ 30% attenuation") are not derived analytically from the dynamical model's equations—such derivation would require parameter values calibrated against empirical data, which have not yet been obtained. Rather, these estimates are *order-of-magnitude predictions* derived from the model's qualitative structure (e.g., the existence of a critical transition implies a large effect near threshold) combined with effect sizes reported in the existing literature on rumination and mindfulness (e.g., the moderate-to-large effect sizes in ACT meta-analyses suggest that operator-level interventions explain a substantial fraction of variance). We state them explicitly so that they can serve as concrete, falsifiable benchmarks for future empirical work; we acknowledge that the exact numerical values will require refinement once the model's parameters are calibrated against longitudinal datasets.

Prediction 1: Dissociation between suffering and unhappiness under defusion interventions. Interventions targeting the negative self-determination operator (such as ACT defusion techniques or mindfulness meditation) should reduce self-reported unhappiness *without necessarily reducing self-reported suffering.* Some evidence for this dissociation already exists in the chronic pain literature [84, 122], but the theory calls for systematic testing across diverse conditions.

Operationalisation. Suffering (s) can be measured by the McGill Pain Questionnaire sensory subscale or by visual analogue scales for distress intensity. Unhappiness (u) can be measured by the Satisfaction With Life Scale (SWLS; 29) or the Oxford Happiness Inventory. The prediction is that a defusion intervention will produce a statistically significant reduction in SWLS-measured unhappiness (Cohen's $d \geq 0.3$) with no significant change in sensory-subscale pain scores.

Prediction 2: Rumination criticality and depression onset. The dynamical-systems model predicts a critical transition in depression onset governed by the ratio of the rumination parameter δ to the operator decay rate β.

Operationalisation. The rumination parameter δ maps onto the brooding subscale of the Ruminative Response Scale (RRS; 119). The operator decay rate β maps inversely onto trait rumination duration, measurable via experience-sampling methodology (ESM). The stressor intensity f_0 maps onto the Life Events and Difficulties Schedule (LEDS; 20). The model predicts that the product $\text{RRS}_{\text{brooding}} \times \text{LEDS}_{\text{severity}}$ should exhibit a nonlinear (threshold-like) relationship with depression onset as measured by the PHQ-9, with the threshold occurring at a characteristic value that can be estimated from longitudinal data. Specifically, the model predicts that individuals above the 75th percentile on RRS-brooding should transition to clinical depression (PHQ-9 $\geq$ 10) at stressor intensities approximately 40–60% lower than those below the 25th percentile.

Prediction 3: Meta-cognitive awareness as a moderator. Individuals trained in meta-cognitive awareness—who can observe the operator $\mathcal{I}^-$ without engaging it—should show a reduced correlation between life adversity and unhappiness.

Operationalisation. Meta-cognitive awareness maps onto the Experiences Questionnaire decentring subscale [42] and the Five Facet Mindfulness Questionnaire (FFMQ) non-reactivity facet [12]. The prediction is that the regression coefficient relating LEDS stressor scores to SWLS scores should be significantly attenuated (by $\geq$ 30%) in participants scoring above the median on decentring, after controlling for neuroticism and baseline well-being.

Prediction 4: Cross-cultural mediation. Cross-cultural differences in unhappiness should be mediated by culturally specific patterns of self-referential evaluation rather than by objective living conditions alone. This is broadly consistent with the Easterlin paradox [33] and could be tested more directly with cross-cultural ESM methods, predicting that between-culture variance in momentary unhappiness ratings is reduced by $\geq$ 40% when RRS-brooding scores are included as a mediator.

6.2 Boundary Conditions and Limitations

Several important boundary conditions must be acknowledged.

The theory does *not* apply to non-conscious suffering. Infants, individuals with severe cognitive impairment, and non-human animals may experience first-order distress without the meta-cognitive capacity for self-referential evaluation. Whether such beings can be "unhappy" in the sense defined here is a philosophical question that our framework addresses by stipulation: unhappiness, as we define it, is a meta-cognitive state and therefore requires the capacity for self-referential thought.

The theory does not deny the reality or moral significance of first-order suffering. A person in severe pain who does not identify as "unhappy" is still in pain, and that pain still makes moral claims on others. The framework separates ontological analysis from moral evaluation.

The binary operator model is an idealisation. In practice, the operator admits of degrees (as captured by the continuous formulation) and is influenced by a host of contextual factors. The dynamical-systems model, while illuminating, involves simplifications (e.g., linearity in the decay terms, two-variable coupling) that may not capture all the complexity of real-world affective dynamics.

Finally, the claim that negative self-determination is a *principal* source of unhappiness—rather than one among many co-equal factors—is a strong theoretical commitment. We do not assert that neurochemical, pharmacological, or neurological processes can never produce negative affective states independently of self-referential cognition; cases such as anhedonia in Parkinson's disease or pharmacologically induced dysphoria represent genuine boundary conditions. Our claim is that in the vast majority of cases encountered in everyday human experience, the transition from suffering to unhappiness is mediated by the operator formalised here. The strength of the claim lies in its capacity to organise a wide range of phenomena and to generate novel predictions; as with any theoretical commitment, it invites rigorous testing and possible refinement.

7 Implications

7.1 For Therapeutic Practice

The framework provides a unified rationale for diverse therapeutic approaches. Cognitive-behavioural therapy targets the content of the negative verdict ($\mathcal{V}^-$). Mindfulness-based interventions target the reflexive turn toward self-evaluation ($\mathcal{R}$). Acceptance and commitment therapy targets the entire operator ($\mathcal{I}^-$) by promoting defusion and psychological flexibility. The dynamical-systems model suggests that reducing the rumination parameter δ may be the most high-leverage intervention, as it prevents the self-reinforcing loop that leads to clinical depression.

7.2 For Public Health and Policy

If unhappiness is generated by a cognitive operator rather than by objective conditions alone, then public-health interventions aimed at improving subjective well-being should target not only material conditions but also the population-level prevalence of negative self-determination patterns. School-based mindfulness programmes [131], resilience training for healthcare workers [98], and cognitive-behavioural public health campaigns could be understood as efforts to reduce the mean activation of $\mathcal{I}^-$ across the population.

This implication must be handled with care. It should *not* be read as a justification for neglecting material deprivation on the grounds that "happiness is all in the mind." Material deprivation increases the input function $f(t)$ and thereby raises the probability of operator activation. The correct policy implication is that *both* material improvement and cognitive-skill development are important, and that neglecting either is suboptimal.

7.3 For Philosophical Anthropology

The framework implies that unhappiness is a distinctly human (or at least meta-cognitive) phenomenon. It arises from the capacity for self-referential thought—the ability to take oneself as an object of evaluation. This capacity is both the source of human dignity (we can reflect, judge, and choose) and the source of human unhappiness (we can condemn ourselves). The theory thus contributes to the philosophical understanding of the human condition by identifying a precise mechanism through which the gift of self-awareness becomes a burden.

7.4 Strategies of Operator Deactivation: Horizon Expansion and Self-Abstraction

The formal framework identifies the negative self-determination operator $\mathcal{I}^-$ as the generative mechanism of unhappiness. A natural question follows: by what means can this operator be deactivated or prevented from firing? The literature and the history of human thought suggest three principal strategies, each of which operates by altering the *scope of reference* within which the subject situates its experience. We term these collectively *strategies of horizon expansion*, because each functions by enlarging the interpretive frame beyond the boundaries of the suffering self.

7.4.1 The Religious Strategy: Ontological Dualism and the Two-World Frame

The oldest and most widespread strategy is the religious one. Across traditions—Christianity, Islam, Hinduism, Buddhism, Judaism—the subject is invited to interpret earthly suffering within a frame that extends beyond the present life. The Christian doctrine of resurrection, the Islamic concept of *ākhirah* (the hereafter), the Hindu and Buddhist frameworks of *saṃsāra* and *mokṣa* or *nirvāṇa*, and the Judaic notion of *olam ha-ba* (the world to come) all perform the same structural operation: they posit a second ontological domain—an afterlife, a higher reality, a transcendent plane—against which the suffering of the present world is relativised [55, 114, 118].

In the terms of the present framework, the religious strategy deactivates $\mathcal{I}^-$ by expanding the cognitive model of self M to include a trans-temporal, trans-worldly dimension. The evaluative judgment operator $J(A, M)$ yields a low value because the model M against which the affective reaction A is evaluated is no longer confined to the present life. A local suffering $A_{\text{local}}(t)$ cannot produce the conclusion "my life is a failure" when "my life" is understood to extend beyond death into a domain where suffering is redeemed, compensated, or transcended.

Formally, we can represent this as a modification of the scope parameter in the cognitive model:

$$M_{\text{religious}} = M_{\text{secular}} \cup M_{\text{transcendent}}$$

where $M_{\text{transcendent}}$ contains beliefs about post-mortem existence, divine justice, or ultimate spiritual liberation. The hypergeneralisation error (Definition 3.14) is blocked because the "whole" against which the part is measured has been vastly enlarged. The present suffering, however intense, is a finite quantity measured against an infinite or qualitatively different background.

Empirical research on religiosity and well-being provides indirect support for this mechanism. Meta-analyses consistently report a modest positive association between religious involvement and subjective well-being [62, 123]. Pargament's work on religious coping [96] identifies "benevolent religious reappraisal"—interpreting a stressor as part of a divine plan—as one of the most effective coping strategies, precisely because it prevents the local-to-global negative generalisation.

7.4.2 The Philosophical Strategy: Rational Reframing and Conceptual Distance

The philosophical strategy achieves a similar structural effect through reason rather than faith. The Stoic tradition, as discussed in Section 2.1, teaches that suffering belongs to the domain of "things not up to us" (*ta ouk eph' hēmin*) and that the only proper object of concern is one's own rational agency [77, Epictetus]. By redefining the self as primarily a rational agent rather than a suffering body, the Stoic contracts the domain of M to which the operator $\mathcal{I}^-$ can attach.

Existentialist philosophy offers a different variant. Frankl [41], writing from the experience of the concentration camps, proposed that meaning can be found even in unavoidable suffering—a thesis that directly implies the possibility of the suffering–unhappiness gap. Camus [24] argued that the absurdity of existence, once accepted rather than resisted, ceases to produce the despair that results from unfulfilled expectations. Nietzsche's concept of *amor fati*—the love of one's fate, including its painful elements—represents the most radical philosophical deactivation of the operator: not merely the absence of a negative verdict but the affirmation of the totality of experience [89].

In all these cases, the philosophical strategy works by introducing *conceptual distance* between the experiencing self and the evaluating self. The subject learns to observe its own suffering as a philosopher observes a phenomenon: with interest, perhaps with compassion, but without the reflexive identification that triggers $\mathcal{I}^-$. This is the cognitive operation that Schopenhauer called "aesthetic contemplation" [106]—the temporary suspension of the will and its demands, in which the subject ceases to be a wanting, suffering agent and becomes a pure knowing subject.

7.4.3 The Strategy of Self-Abstraction: De-Centring the "I"

The third strategy is the most radical and the most directly targeted at the operator itself. Rather than expanding the horizon (as in the religious strategy) or introducing conceptual distance (as in the philosophical strategy), the strategy of self-abstraction *dissolves the referent* of the operator. If the negative self-determination operator requires a self to determine, then weakening or suspending the self-representation eliminates the substrate on which the operator acts.

This strategy is most fully developed in Buddhist meditative practice, particularly in the *anattā* (non-self) doctrine [44, 49]. Vipassanā meditation trains the practitioner to observe sensations, thoughts, and emotions as impersonal processes—"there is pain" rather than "I am in pain"—thereby disabling the reflexive self-attribution step $\mathcal{R}$ (Definition 4.4). Advanced practitioners report a stable capacity to experience intense sensations without generating self-referential narratives [5, 17].

Contemporary psychological science converges on a similar mechanism through different terminology. The concept of "decentring" in mindfulness-based cognitive therapy [42, 107] describes the capacity to observe one's thoughts and feelings as transient mental events rather than as truths about the self. Kross and colleagues [63] showed that adopting a self-distanced perspective (using one's own name or third-person pronouns rather than "I" when reflecting on negative experiences) reduces emotional reactivity and rumination. These findings suggest that the self-referential pronoun "I" is not merely a linguistic convenience but a cognitive anchor that facilitates the activation of the negative self-determination operator.

In formal terms, the strategy of self-abstraction modifies the reflexive attention operator $\mathcal{R}$ such that:

$$\mathcal{R}_{\text{abstracted}}(S) \to S_{\text{observed}} \quad \text{rather than} \quad S_{\text{self-attributed}}$$

When the first-order state is observed rather than self-attributed, it enters consciousness as a datum—something that is happening—rather than as a verdict—something that defines who I am. The downstream evaluation $\mathcal{V}^-$ has no self-referent on which to operate, and the operator $\mathcal{I}^- = \mathcal{V}^- \circ \mathcal{R}$ returns zero.

7.4.4 Comparative Structure of the Three Strategies

The three strategies can be summarised in terms of their formal operation on the components of the model.

The religious strategy expands M so that the ratio of local suffering to total life-scope approaches zero. The philosophical strategy introduces a secondary evaluative frame that overrides or inhibits $\mathcal{V}^-$. The strategy of self-abstraction disables $\mathcal{R}$ by dissolving the self-reference on which the operator depends.

All three achieve the same functional outcome—the deactivation of $\mathcal{I}^-$—but through different mechanisms and at different points in the processing chain. This structural analysis suggests that the three strategies are not competitors but complementary interventions, and that their effectiveness may vary across individuals depending on temperament, cultural background, and the nature of the suffering encountered. A subject facing terminal illness may find the religious strategy most effective; a subject facing social humiliation may benefit more from philosophical reframing; a subject prone to chronic rumination may respond best to the meditative dissolution of self-reference.

The key insight is that all three strategies, despite their vast differences in metaphysical commitments and cultural provenance, converge on a single functional goal: preventing the transition from first-order suffering to second-order unhappiness by disrupting the conditions under which the negative self-determination operator can fire.

8 Situating the Present Work within a Broader Research Programme

The theory presented in this paper is part of a broader research programme comprising publications across multiple domains (1999–2026) that investigates structural invariants of complex systems [64]. The programme's central thesis is that *persistence* is the primary organising principle of complex systems: truth-seeking, consciousness, and social organisation are derivative strategies for maintaining structural viability.

For the present paper, three results from the programme are directly relevant. First, the Viability Mismatch Law [67] formalises stress in any system as a state in which environmental demands exceed the system's current viability set. Unhappiness, as modelled here, is a cognitive instance of viability mismatch: the subject's predictive model registers a discrepancy that is then amplified by the self-referential operator. Second, the

Structural Distortion Principle [66] establishes that bounded cognitive systems must distort their representations of reality—cognitive biases are optimal solutions under resource constraints, not failures. The rumination loop ($\delta > 0$) can be understood as a distortion that was once adaptive (rapid threat generalisation) but becomes maladaptive when applied to symbolic stressors. Third, the Reflexive Inference Law [65] bounds what any inferential system can validly generalise from self-observation, providing a formal basis for the identification error (Definition 3.13): the hypergeneralisation patterns exceed the scope permitted by the law.

We note explicitly that these connections are presently *structural analogies* rather than formal derivations. A complete derivation of the dynamical model's equations from the programme's axioms has not yet been achieved and remains a goal for future work. The present paper stands on its own merits independently of the broader programme; the connections are noted here to indicate the direction of ongoing research.

9 Conclusion

We have presented a formal theory of unhappiness grounded in a single thesis: *the act of negative self-determination is a principal source of unhappiness*. The theory distinguishes between suffering—a first-order experiential fact—and unhappiness—a second-order, reflexive cognitive act in which the subject identifies itself as defeated. We formalised this distinction using an operator-theoretic framework, stated and established four main propositions, and embedded the theory in a dynamical-systems model that generates testable predictions about the temporal evolution of unhappiness and the conditions for critical transitions into depressive states.

The literature review demonstrated broad convergence across Stoic philosophy, Buddhist psychology, cognitive appraisal theory, predictive processing neuroscience, acceptance and commitment therapy, rumination research, and cross-cultural well-being studies. This convergence lends credibility to the theory while also highlighting the need for further empirical work, particularly in testing the dissociation between suffering and unhappiness and in measuring the rumination criticality threshold.

The theory carries implications for therapeutic practice (targeting the negative self-determination operator), public health (combining material improvement with cognitive-skill development), and philosophical anthropology (understanding unhappiness as a consequence of the capacity for self-referential thought).

We close with a remark on the pragmatic and epistemic status of negative self-determination, followed by the concise formulation that motivates the entire paper.

Remark 9.1 (The Pragmatic and Epistemic Status of Negative Self-Determination)**.** *The act of declaring oneself unhappy can be evaluated on two independent grounds. On* pragmatic *grounds, it is inert: the declaration does not remove the cause of suffering, does not alleviate pain, and does not alter the adverse circumstances. It is an operation with zero instrumental return. On* contaminative *grounds, it is actively harmful: once the verdict "I am unhappy" is rendered, it propagates through the subject's interpretive field and colours experiences that are, in themselves, neutral or even positive. This is the self-reinforcing feedback captured by the parameter* δ *in the dynamical model (Section 5).*

An important objection must be considered. From the standpoint of bounded rationality [46], hypergeneralisation may be a computationally cheap heuristic that is adaptive under certain ecological conditions. If threats are persistent and correlated, then generalising from a single negative signal to a global negative state may be a prudent "better safe than sorry" strategy that served ancestral populations well. We do not dismiss this possibility. However, we note two qualifications. First, the ecological conditions under which the heuristic was adaptive (persistent, correlated, life-threatening dangers in small-scale societies) differ markedly from the conditions of contemporary life, where the majority of stressors are symbolic, social, and chronic rather than acute and lethal. A heuristic that was adaptive on the savanna may be systematically maladaptive in an office. Second, even if the heuristic was once adaptive at the population level, it does not follow that it is rational at the individual level in the present. The distinction between ecological rationality (fitness-enhancing on average) and epistemic rationality (truth-tracking for the individual) is well-established [116].

With these qualifications, we characterise negative self-determination not as "foolish" in a dismissive sense but as a scope error*—a judgment whose conclusion exceeds its evidence (Definition 3.16). To the extent that the error can be recognised and interrupted, the subject gains a degree of freedom that would otherwise be foreclosed.*

The eradication of suffering is a noble but perhaps unattainable goal. The disengagement of the operator that transforms suffering into unhappiness is, by contrast, a definite and practicable aim—one toward which philosophy, psychotherapy, and contemplative practice have always, in their different ways, been directed.

The paper's central thesis can be distilled into two complementary formulations:

> *Suffering is a fact. Unhappiness is the interpretation of suffering as the defeat of the self.*
>
> *To consider oneself unhappy changes nothing; it only casts negativity over everything.*

The first formulation is ontological: it locates the origin of unhappiness in a specific cognitive act. The second is pragmatic: it observes that the act is instrumentally futile and contaminatively harmful. Together, they constitute both the theoretical core and the practical conclusion of the present work.

A Case Studies: Dual-Pathway Simulations

The following case studies illustrate the theory's central distinction by simulating the four-variable dynamical model of Section 5 (Equations 1–4, with the memory equation 3 and all saturation nonlinearities) under two contrasting parameter regimes for the same stressor event. In each case, **Path A** represents a subject in whom the negative self-determination operator fires (high rumination δ, high memory coupling κ, low operator decay β), while **Path B** represents a subject in whom the operator is not engaged or is quickly extinguished (low δ, low κ, high β). The first-order suffering input $f(t)$ is identical in both paths. All state variables are confined to $[0, 1]$ by the saturating nonlinearities in the equations; no post-hoc capping is applied.

Each figure displays the three state variables—suffering $s(t)$, operator activation $a(t)$, and unhappiness $u(t)$—evolving over time. The contrast between the two rows within each figure demonstrates the paper's core thesis: identical first-order suffering can produce dramatically different levels of unhappiness depending on whether the negative self-determination operator is activated.

A.1 Case Study 1: Job Loss

Event. A 42-year-old professional is unexpectedly terminated from a long-held position. The stressor produces a sharp initial spike of distress ($f_0 = 0.8$) that decays exponentially as the acute shock subsides.

Path A ($\delta = 1.8$, $\beta = 0.2$, $\gamma = 0.9$): The subject interprets the termination as confirmation of personal inadequacy. The thought "I am a failure" engages the rumination loop. Operator activation rises rapidly and remains elevated; unhappiness saturates near its maximum and persists long after the first-order distress has partially subsided. This trajectory illustrates *identity hypergeneralisation* (Definition 3.14, pattern iii): a discrete event (losing one job) is projected onto the subject's core self-model.

Path B ($\delta = 0.1$, $\beta = 1.2$, $\gamma = 0.3$): The subject registers the loss as a significant but bounded setback. The thought pattern is "This is a blow, but it does not define who I am." Operator activation remains low throughout; unhappiness peaks briefly and decays quickly. The suffering–unhappiness gap (Definition 3.9) is clearly visible: substantial suffering with minimal unhappiness.

A.2 Case Study 2: Chronic Pain Diagnosis

Event. A patient receives a diagnosis of a chronic pain condition. Unlike the previous case, the stressor is *persistent*: the input function $f(t)$ does not decay to zero but oscillates around a sustained baseline ($f_0 \approx 0.5$).

Path A ($\delta = 2.0$, $\beta = 0.15$, $\gamma = 0.7$): The patient forms the judgment "My life is

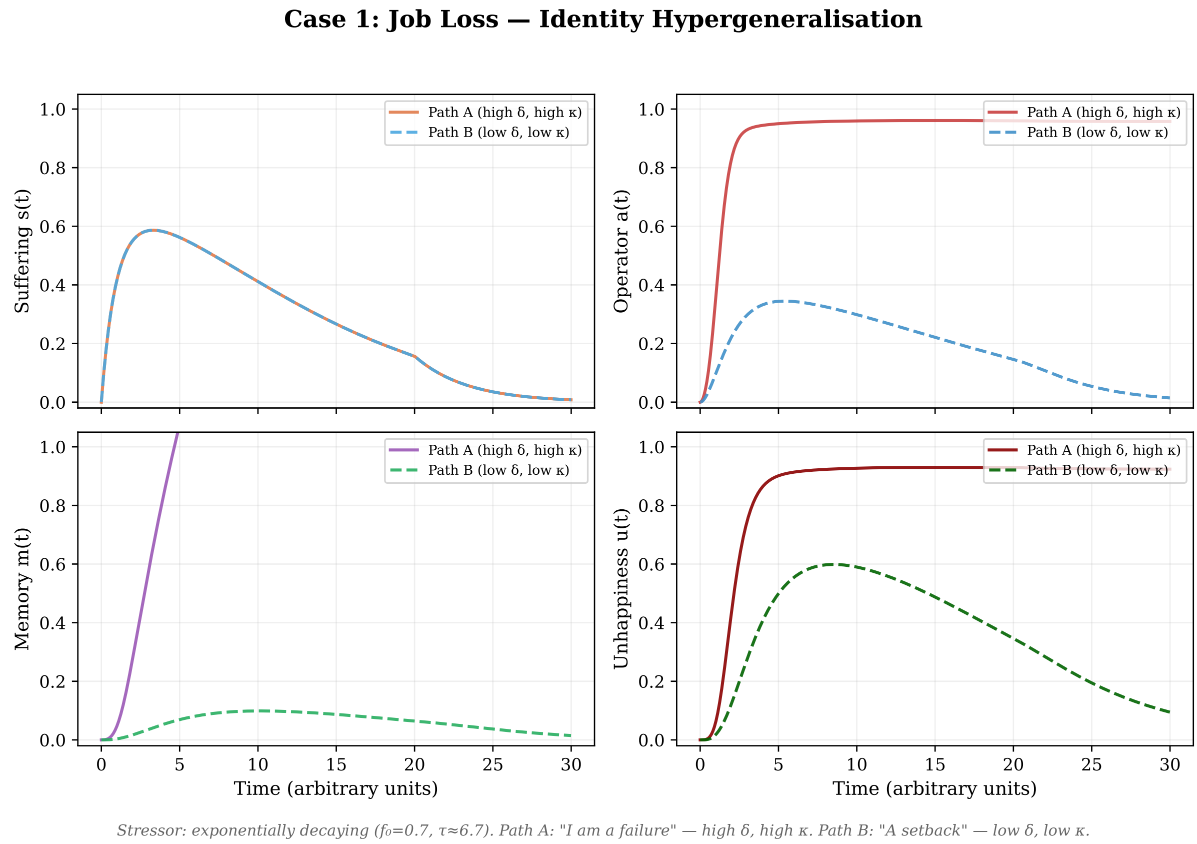

Figure 1: Case Study 1: Job loss. Top row: Path A (operator fires, $\delta = 1.8$). Bottom row: Path B (operator extinguished, $\delta = 0.1$). The stressor is identical; unhappiness diverges due to operator engagement.

ruined." The combination of persistent input and high rumination drives the operator to its upper bound. Unhappiness becomes chronic and maximal. This case illustrates *temporal hypergeneralisation* (Definition 3.14, pattern i): a present state ("I am in pain now") is projected indefinitely into the future ("I will always be in pain and my life will always be miserable").

Path B ($\delta = 0.05$, $\beta = 0.8$, $\gamma = 0.2$): The patient, perhaps through an acceptance and commitment therapy (ACT) programme, learns to experience pain without forming a self-referential verdict. The thought pattern is "There is pain" rather than "I am a person in unending misery." Operator activation remains near zero; unhappiness is minimal despite sustained, significant suffering. This is the chronic-pain acceptance pattern documented by McCracken [84] and Veehof and colleagues [122].

A.3 Case Study 3: Bereavement

Event. The death of a spouse produces the most intense stressor in the set ($f_0 = 0.95$), with a slow exponential decay modulated by periodic grief surges (anniversary reactions, intrusive memories).

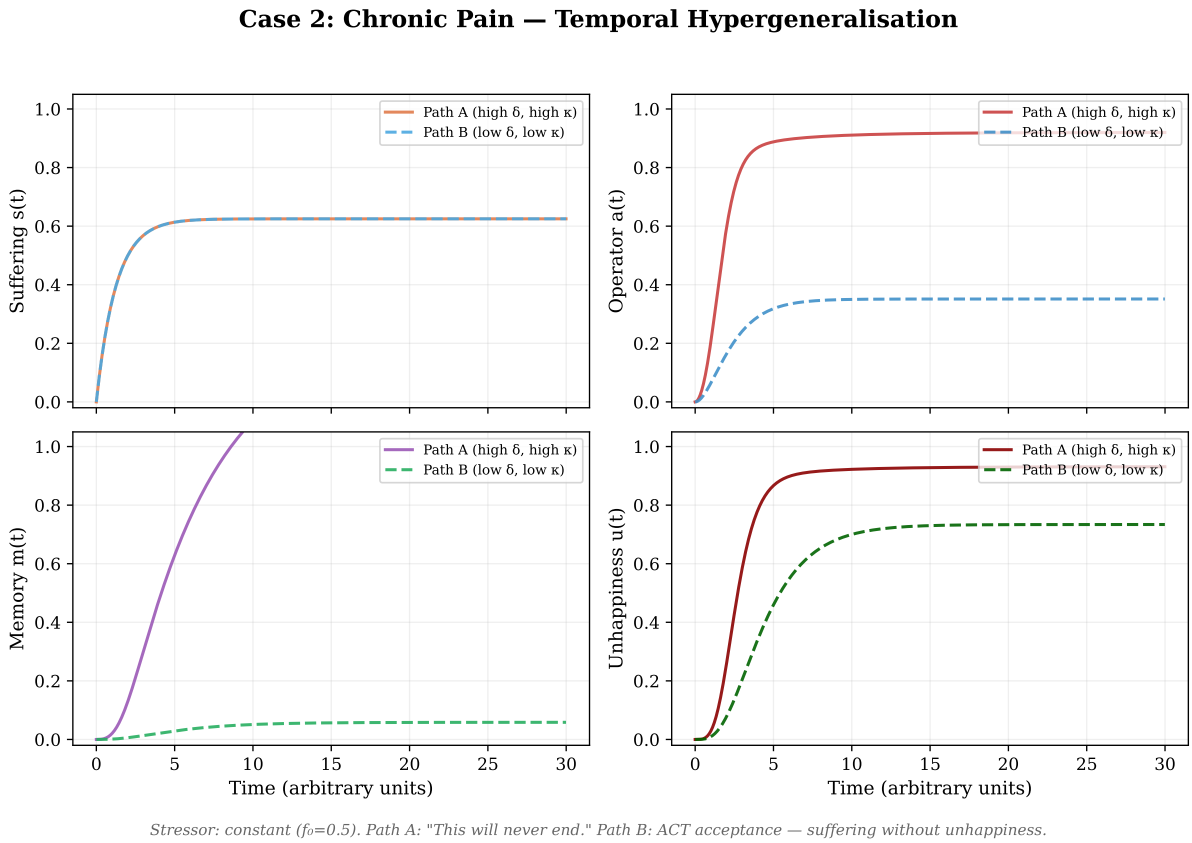

Figure 2: Case Study 2: Chronic pain. The stressor is persistent and does not decay. Path A (top) shows runaway operator activation under high rumination. Path B (bottom) shows acceptance: suffering is sustained but unhappiness remains near zero.

Path A ($\delta = 2.5$, $\beta = 0.12$, $\gamma = 0.8$): The bereaved subject forms the conviction "I will never recover; my life is over." The extremely high rumination parameter, combined with slow operator decay and intense input, drives the system toward its critical threshold. Unhappiness remains near maximum over the full simulation horizon. This trajectory is consistent with prolonged grief disorder (ICD-11), in which self-referential rumination prevents the natural resolution of bereavement [92].

Path B ($\delta = 0.1$, $\beta = 0.6$, $\gamma = 0.25$): The bereaved subject grieves deeply—the suffering curve is indistinguishable from Path A in the early phase—but does not form a totalising self-verdict. The grief is experienced as an expression of love, not as a judgment of defeat. Meaning-making, as described by Frankl [41], holds the operator in check. Unhappiness peaks and then resolves over time even though suffering remains substantial.

A.4 Case Study 4: Academic Failure

Event. A graduate student fails a critical qualifying examination. The stressor is brief and intense ($f_0 = 0.7$, duration ≈ 0.5 time units), modelling the acute shock of receiving

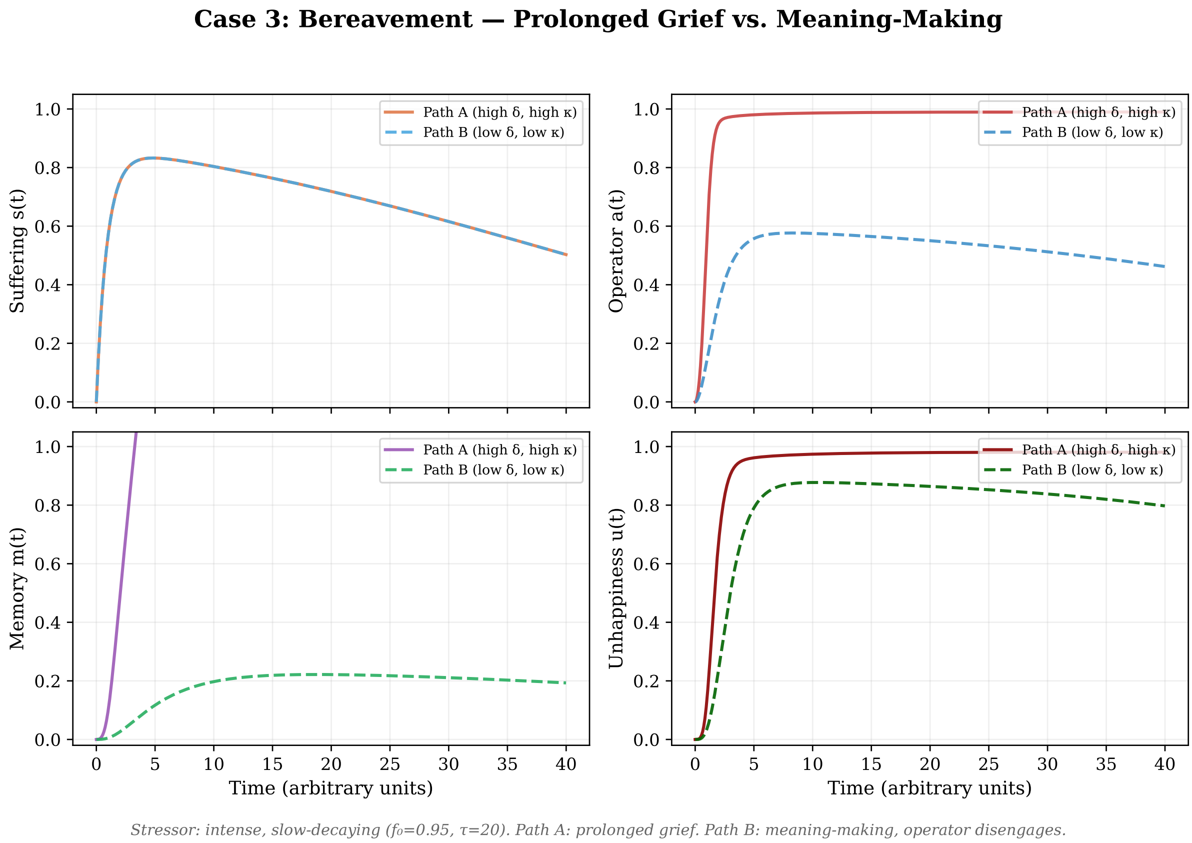

Figure 3: Case Study 3: Bereavement. The most intense stressor in the set. Path A (top) shows prolonged grief with maximal operator activation. Path B (bottom) shows adaptive grief: intense suffering that does not produce sustained unhappiness.

the result.

Path A ($\delta = 2.0$, $\beta = 0.2$, $\gamma = 1.0$): The student concludes "I am stupid"—a classic instance of identity hypergeneralisation. Despite the brevity of the stressor, the strong coupling and high rumination sustain operator activation well after the first-order distress has dissipated. Unhappiness outlasts suffering by a wide margin. This dissociation—transient suffering, prolonged unhappiness—is a signature prediction of the model: the operator can "store" the negative verdict and sustain unhappiness independently of the original input.

Path B ($\delta = 0.1$, $\beta = 1.0$, $\gamma = 0.3$): The student registers the failure as a discrete event: "I failed this test. I will prepare differently." The operator barely activates. Both suffering and unhappiness resolve rapidly. The difference between the two paths is not in the event but in the scope of the judgment.

A.5 Case Study 5: Repeated Social Rejection

Event. Three episodes of social rejection occur at $t = 3$, $t = 8$, and $t = 14$, with decreasing intensity ($f_0 = 0.6, 0.4, 0.3$). This case tests the model's behaviour under

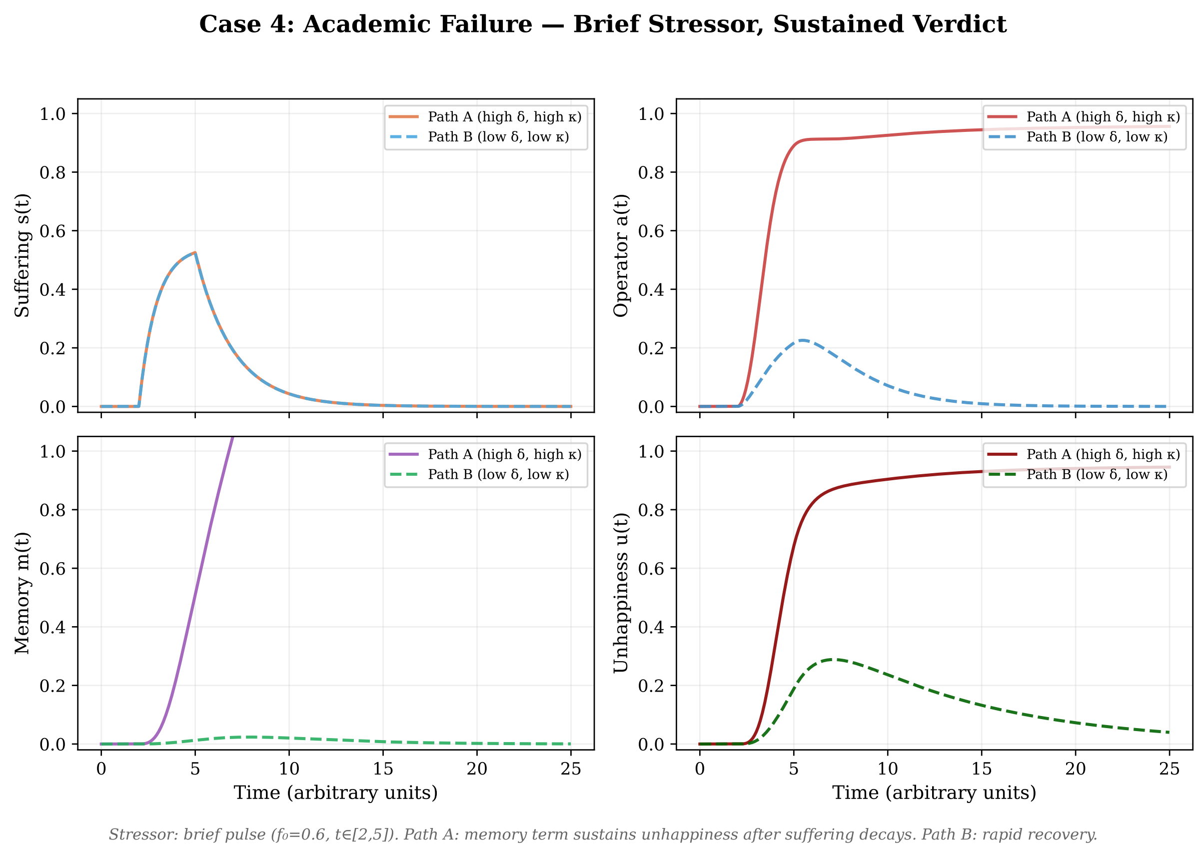

Figure 4: Case Study 4: Academic failure. A brief, intense stressor. In Path A (top), the operator sustains unhappiness long after suffering has decayed—demonstrating that unhappiness can outlast its triggering event. In Path B (bottom), both resolve rapidly.

repeated, spaced stressors.

Path A ($\delta = 2.2$, $\beta = 0.15$, $\gamma = 0.8$): Each rejection episode reactivates and amplifies the operator. The narrative "Nobody likes me" generalises across episodes (*scope hypergeneralisation*). Crucially, the third rejection—objectively the mildest—produces the largest unhappiness spike, because the operator is primed by residual activation from the first two episodes. This is the rumination accumulation effect, consistent with Proposition 5.1: cumulative activation pushes the system closer to the critical threshold with each event.

Path B ($\delta = 0.05$, $\beta = 1.0$, $\gamma = 0.25$): Each episode produces a brief, proportionate distress response that resolves before the next episode occurs. The operator does not accumulate across episodes. The third rejection produces less distress than the first, consistent with habituation and bounded processing.

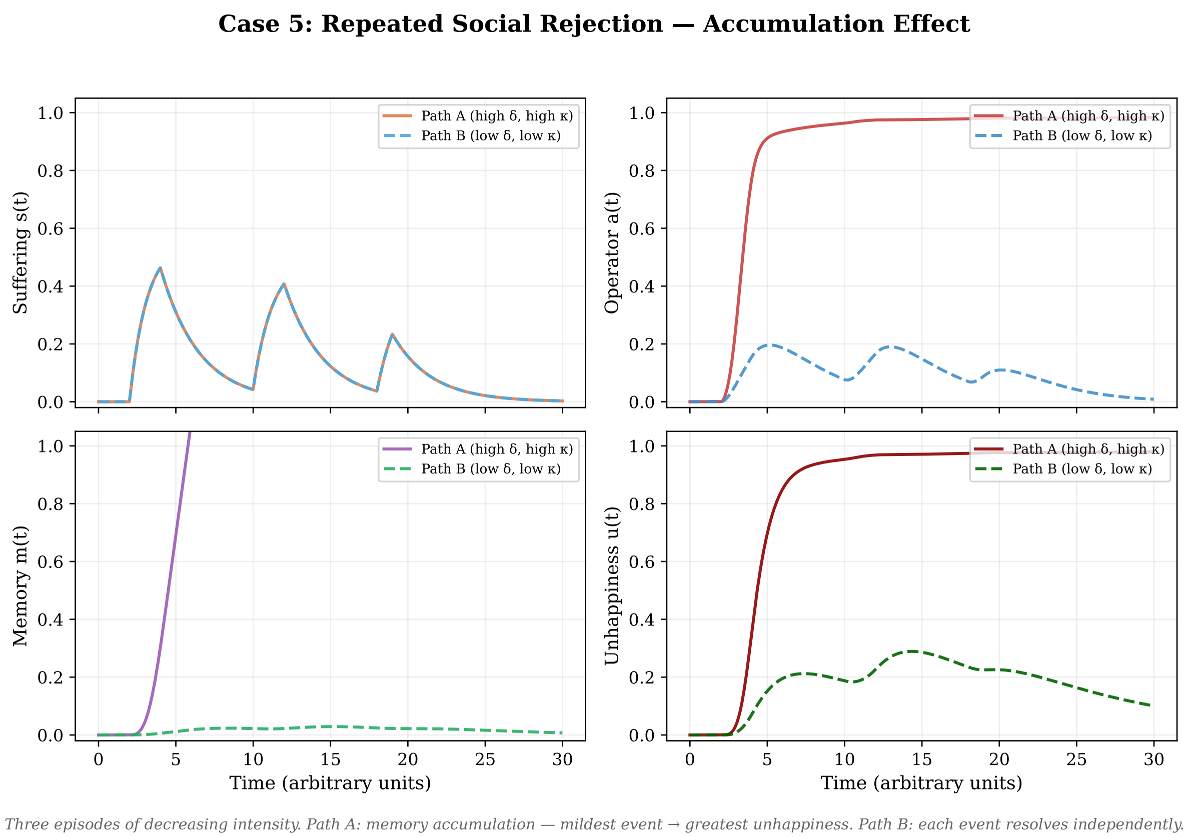

Figure 5: Case Study 5: Repeated social rejection. Three episodes of decreasing objective severity. In Path A (top), the operator accumulates across episodes via the memory term $m(t)$: the mildest event produces the greatest unhappiness. In Path B (bottom), each episode is processed independently.

A.6 Case Study 6: Existential Depression — Unhappiness Without Proportionate Cause

This case directly addresses the reconciliation of the dynamical model with Proposition 4.2 (sufficiency of the operator). The subject lives in objectively comfortable circumstances: $f(t) = 0.05$ throughout (minimal background stressor). First-order suffering $s(t)$ remains near zero.

In Path A, the subject has high memory coupling ($\kappa = 0.8$), high memory deposition rate ($\eta = 0.6$), slow memory decay ($\nu = 0.08$), and strong rumination ($\delta = 5.0$). Despite near-zero current suffering, the memory term $m(t)$ accumulates from even minimal operator activation: anticipated suffering, recalled distress, and existential preoccupations ($m > 0$) feed back into the operator through the κm term in Equation (2). The operator ramps up, deposits more memory, and a self-sustaining loop emerges. The result is substantial unhappiness ($u \to 0.4$–0.6) with negligible current suffering ($s \approx 0.05$)—the suffering–unhappiness gap operating in reverse.

In Path B, with low κ, low η, and low δ, the same negligible stressor produces negligible operator activation and no unhappiness.

This case demonstrates that the memory term $m(t)$ introduced in Equation (3) is not merely a technical convenience but a substantive extension that captures a clinically recognised phenomenon: rumination-driven depression in the absence of proportionate external adversity. It models the clinical presentations often described as "endogenous depression" or "existential crisis," where the patient reports unhappiness that is not attributable to any identifiable stressor.

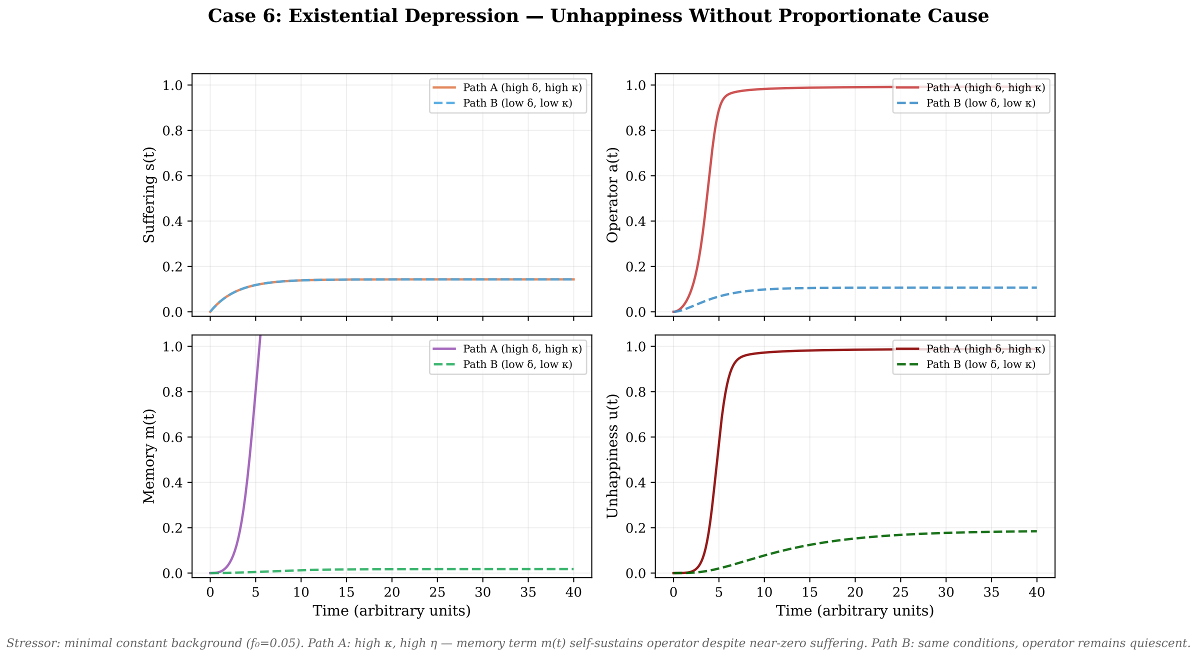

Figure 6: Case Study 6: Existential depression. Minimal stressor ($f_0 = 0.05$) throughout. In Path A, the memory term $m(t)$ self-sustains the operator: unhappiness emerges without proportionate suffering, reconciling the dynamical model with Proposition 4.2. In Path B, the same negligible stressor produces no operator activation.

A.7 Parameter Calibration Note

The parameter values used in all six case studies were chosen to illustrate the qualitative dynamics predicted by the model (dual pathways, critical transitions, suffering–unhappiness gaps, memory-sustained unhappiness) rather than derived from or calibrated against empirical data. They demonstrate *internal consistency*—the model produces the predicted qualitative patterns—but not *empirical validity* in the quantitative sense. Full parameter calibration would require fitting the model to longitudinal experience-sampling data in which momentary suffering, rumination, and unhappiness are measured concurrently across diverse stressor types. Such a study is a priority for future work.

A.8 Summary: The Criticality Landscape

Figure 7 summarises the relationship between the rumination parameter δ and the steady-state operator activation a^* for three levels of stressor intensity, computed numerically from the saturating model (Equation 6). As δ increases, the equilibrium a^* transitions from a low-activation regime to a high-activation regime via a saddle-node bifurcation (Proposition 5.1). Higher stressor intensity shifts the critical threshold to the left, meaning that less rumination is required to trigger the transition.

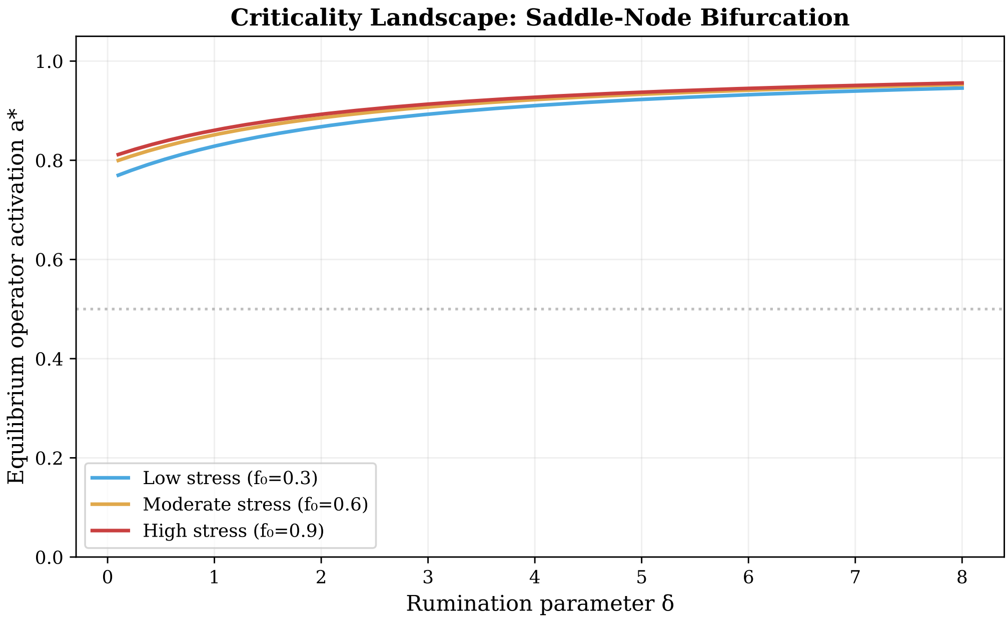

Figure 7: The criticality landscape (saturating model). Steady-state operator activation a^* as a function of rumination parameter δ for three levels of stressor intensity f_0. The transition from low to high activation occurs via a saddle-node bifurcation rather than a pole, ensuring that the model is well-defined throughout. Higher stress shifts the critical threshold leftward.

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